Hair-on-Fire Demand Map

Proposal for Mike — Oct 31, 2026 (America/New_York).
Hair-on-fire pains ranked by urgency × willingness to pay × MIRA fit, plus a verified public-complaint survey (45 counted instances with linked proof).

Info only. Nothing here is customer-facing ship work. No public prices. This map does not authorize merge, deploy, Doppler, networking, PLC, paid cloud agents, or customer-facing launch.

Source Page: Hair-on-Fire Demand Map

How to read this

Scores are Foreman judgment from evidence density and MIRA Capability Closure — not invented survey percentages.

Rank = urgency (hair-on-fire now) × willingness to pay (budget already moving) × MIRA fit (reuse the sellable SUT: mira-mobile → Hub equipment-notebook chat → manual RAG → citation SSE).

Status tags: REUSE what exists, CONNECT manuals/sources, FINISH near-done capabilities, BUILD only when nothing closer wins.

Sellable SUT	mira-mobile → Hub equipment-notebook chat → `retrieveNodeChunks` → citation SSE
Positioning	MIRA stands alone; CMMS connectors (Atlas etc.) are optional add-ons
Repo snapshot	Mikecranesync/MIRA — CAPABILITY_CLOSURE: notebook turns + approved retrieval + `pilot_access` `production_enabled`; `guided_troubleshooting` / nameplate / photo OCR `deployed_disabled`

Priority scoreboard

#	Pain	Urgency	Pay	Fit	Score	Status
1	Cited answers from OUR manuals at the machine	10	9	10	9.7	FINISH
2	Junior / night shift as capable as the retiring senior	9	9	9	9.0	FINISH
3	Cut recovery time when unplanned downtime hits	10	10	8	9.3	REUSE
4	Branching troubleshooting that isn't a useless error-code echo	8	8	9	8.3	FINISH
5	Photo the nameplate → right manual → ask	7	7	9	7.7	FINISH

#	Pain	Urgency	Pay	Fit	Score	Status
6	AI that works without ripping out (or depending on) the CMMS	7	8	9	8.0	REUSE
7	Record the fix once; next shift gets the guided workflow	7	7	7	7.0	CONNECT
8	Answers when the help file needs internet (or the cell drops)	6	6	6	6.0	BUILD

Top three to build and market now: cited manual Q&A, junior/night-shift competence, and MTTR recovery (same product sold to ops).

Upper-right product lane: #1 cited manuals, #2 tribal knowledge, #3 MTTR (reuse #1), #4 guided fault trees, #6 standalone wedge. Offline (#8) stays lower until pilots prove online value.

Demand detail

#1 — Cited answers from OUR manuals at the machine

- **Status:** FINISH
- **Buyer:** Maintenance manager + plant ops (MTTR buyers)
- **Evidence:** MaintainX 2026: 28% cite skills gap and 36% labor shortage as downtime drivers; r/PLC techs already upload OEM manuals into NotebookLM/ChatGPT because generic AI hallucinates; Squint, Siemens Asset Essentials Copilot, and Rockwell's Singapore Maintenance Copilot all sell this exact job; Rockwell reports 33% lower downtime and onboarding 9→3 months at that site.
- **MIRA status:** Core SUT exists: mira-mobile → Hub equipment-notebook chat → `retrieveNodeChunks` (manual-rag) → citation SSE. Capability Closure: `private_notebook_turns` + `approved_context_retrieval` + `pilot_access` = `production_enabled`; `notebook_manual_acquisition` = `staging_enabled`. Answer Radar freezes keep parking because plant-specific manuals are not CONNECTED into notebook `sourceDocIds` yet.
- **Smallest sellable move:** Ship one sellable pilot SKU: upload customer's top 20 OEM manuals → ask fault/parameter questions on phone → always cite page or honest ABSTAIN. Prove on 3 Answer Radar–class machines (e.g. Rockwell drive + Siemens HMI + one Schneider I/O) before any CMMS story.
- **Marketing hook (draft for Mike — not shipped):** Stop Ctrl+F'ing a 300-page Rockwell PDF while the line is down. Ask in plain English. Get the page. Or we tell you we don't know.

#2 — Junior / night shift as capable as the retiring senior

- **Status:** FINISH
- **Buyer:** Plant director / reliability lead facing retirements
- **Evidence:** Deloitte + Manufacturing Institute (2024): up to 1.9M of 3.8M needed U.S. manufacturing roles could go unfilled through 2033; MaintainX 2026 ranks labor shortage and

poor knowledge transfer among top downtime drivers; Rockwell Singapore plant director names ‘tribal knowledge’ as the problem the Copilot solves; r/IndustrialMaintenance ‘Zero documentation’ thread and PLCtalk ‘Documenting Projects’ show plants living this daily.
- **MIRA status:** Same notebook path amplifies juniors with OEM truth; `guided_troubleshooting` (diagnostic-plan runner + procedure sessions) exists but is `deployed_disabled`; veteran-fix capture into notebook sources is CONNECT work, not a new brain.
- **Smallest sellable move:** Pilot offer: ‘Senior in a pocket’ — bind one plant’s critical assets to manuals + 5 recorded veteran fixes as notebook sources; measure time-to-first-cited-answer for a new hire on night shift.
- **Marketing hook (draft for Mike — not shipped):** Your best tech retires Friday. Their manuals and fixes should still answer the 2am fault Monday.

#3 — Cut recovery time when unplanned downtime hits

- **Status:** REUSE
- **Buyer:** Ops / finance sponsors who feel rising downtime cost
- **Evidence:** MaintainX State of Industrial Maintenance 2026 (n=2,234): 79% saw unplanned downtime stay the same or rise; 39% say downtime cost rose (vs 31% in 2025); half of teams still spend <40% of time on planned work despite PM programs. Buyers fund tools that shorten recovery, not another dashboard.
- **MIRA status:** Do not build a CMMS downtime module. Position MIRA as the recovery accelerator on top of whatever WO system they already have. Atlas/CMMS connectors stay optional add-ons per Mike’s positioning.
- **Smallest sellable move:** Marketing + pilot metric only: median minutes from fault code asked → cited next step, vs hunting paper/PDF. No public dollar claims.
- **Marketing hook (draft for Mike — not shipped):** Downtime isn’t getting cheaper. The fastest recovery path is the one that finds the right page first.

#4 — Branching troubleshooting that isn’t a useless error-code echo

- **Status:** FINISH
- **Buyer:** Controls/maintenance techs burning hours on circular OEM guides
- **Evidence:** r/PLC: Automate troubleshooting books that say ‘problem with breaker x.y.z’ for hundreds of codes; Rockwell manuals described as 300-page Ctrl+F hell with chopped part numbers; Squint markets model-specific steps cited from OEM manuals; MIRA already has manufacturer diagnostic plans (e.g. SMC Flex / 150-UM008J Ch.9).
- **MIRA status:** `guided_troubleshooting` capability = `deployed_disabled` (WP-1 equipment notebook procedure route + diagnostic-plan guided runner). Smallest path: enable for one drive family in pilot, not a general agent.
- **Smallest sellable move:** Finish one OEM chapter → interactive branches in the notebook (measure → decide → next). Keep chat Q&A as the always-on fallback when no tree matches.

- **Marketing hook (draft for Mike — not shipped):** Fault Fxxx shouldn't dump you into a book that restates the fault. Walk the branch. Cite the chapter.

#5 — Photo the nameplate → right manual → ask

- **Status:** FINISH
- **Buyer:** Multi-OEM plants where techs can't name the exact drive/HMI
- **Evidence:** Remote-site r/PLC users prep troubleshooting steps during 2-hour drives with Gemini because they cover 'so many devices and manufacturers'; Answer Radar freezes repeatedly block on missing CONNECTED manuals for specific SKUs (Omron, Schneider, Siemens HMI, Kinetix, etc.). Discovery is half the fire.
- **MIRA status:** `nameplate_identification` + `photo_ocr_evidence` + `manual_discovery_judge` are `deployed_disabled` (pilot deferred per Closure — owner put first pilot on manual-grounded troubleshooting). `notebook_manual_acquisition` is `staging_enabled`.
- **Smallest sellable move:** After cited Q&A is sellable: supervised Pixel dogfood of nameplate → candidate manuals → attach → ask. Do not lead GTM with computer vision.
- **Marketing hook (draft for Mike — not shipped):** Point the phone at the nameplate. Talk to that machine's manual — not the internet.

#6 — AI that works without ripping out (or depending on) the CMMS

- **Status:** REUSE
- **Buyer:** Plants stuck on Fiix/Limble/UpKeep/MaintainX who still Ctrl+F manuals
- **Evidence:** r/IndustrialMaintenance: Fiix 'too complicated for many technicians'; Limble mobile battery/load complaints; MaintainX Assist still lives inside a WO/procedure CMMS; OEM copilots (Siemens Asset Essentials) are locked to their EAM. Buyers want wrench-time answers without a CMMS migration.
- **MIRA status:** Product positioning already locked: MIRA stands alone; CMMS connectors (Atlas etc.) are optional. Honest-landing work stripped auto-WO and free-CMMS claims from public HTML.
- **Smallest sellable move:** Lead every pilot and sales deck with standalone notebook Q&A. Offer Atlas comment sync only after they love cited answers. Never imply MIRA needs a CMMS.
- **Marketing hook (draft for Mike — not shipped):** Keep your CMMS. Add the assistant that actually reads the manuals.

#7 — Record the fix once; next shift gets the guided workflow

- **Status:** CONNECT
- **Buyer:** Reliability leads who already digitize SOPs (Augmentir/Squint buyers)
- **Evidence:** Augmentir / Squint sell capture→guided workflow and remote expert; Inductive Automation Cooley Group case: tribal troubleshooting lost when people leave; MaintainX AI turns manuals/PDFs into procedures. Content authoring labor is the competitor complaint.

- **MIRA status:** CONNECT: treat approved veteran notes / photos as notebook sources beside OEM manuals; reuse guided procedure sessions. Avoid building a full connected-worker authoring suite first.
- **Smallest sellable move:** Pilot add-on after cited OEM Q&A: ‘attach last good fix’ to the asset notebook; cite it like a manual page. Skip AR/remote-expert theater until pull is proven.
- **Marketing hook (draft for Mike — not shipped):** If it took your senior an hour to find, the next tech shouldn’t start from zero.

#8 — Answers when the help file needs internet (or the cell drops)

- **Status:** BUILD
- **Buyer:** Basement MCC / remote plants / OT-isolated networks
- **Evidence:** r/PLC: ‘help files now require an internet connection’; PLCtalk paperless-maintenance thread: techs still print key pages for side-by-side troubleshooting. Offline is real but second to ‘right cited answer when online.’
- **MIRA status:** mira-mobile has offline WO queue work historically; full offline manual RAG + citation viewer is not the sellable SUT. BUILD only after online cited Q&A wins pilots.
- **Smallest sellable move:** Park as post-pilot: cache attached notebook manuals + last N cited answers for airplane/MCC mode. Do not delay GTM for it.
- **Marketing hook (draft for Mike — not shipped):** When the plant network hates you, your manuals shouldn’t vanish too.

Recommended build order

1. **Finish the cited-manual pilot SKU** — Stranger uploads manuals → asks → gets page citations or ABSTAIN. Gate with `beta_ready_upload_retrieval_citation`-class proof on staging + one Pixel path. CONNECT the manuals Answer Radar already named.
2. **Market to the MTTR + skills-gap buyer with the same SKU** — One product, two hooks: ‘line-down recovery’ for ops and ‘senior in a pocket’ for workforce. No CMMS migration pitch. No public price tags.
3. **Enable guided troubleshooting for one OEM family** — Turn on `deployed_disabled guided_troubleshooting` for a single drive chapter already coded (e.g. SMC Flex). Keep chat as fallback.
4. **Optional Atlas after love** — Only once cited answers stick: optional CMMS comment/work-note sync. Never block the standalone sale on it.
5. **Park nameplate CV and offline RAG** — Supervised dogfood later. Do not let BUILD work delay the FINISH/CONNECT pilot.

Competitor / market landscape

Who is already selling the fire

1. Squint — plain-language faults answered from OEM manuals/procedures with page cites; 2am ‘best tech in their pocket’ messaging
2. Siemens Asset Essentials Maintenance Copilot — attached PDF manuals + history; cites document name and page; enterprise EAM-locked
3. Rockwell Singapore GenAI Maintenance Copilot (Microsoft Source, Sep 2026) — manuals + veteran knowledge; claimed 33% downtime cut, onboarding 9→3 months
4. MaintainX Assist — manuals + procedures + work history inside a CMMS
5. Tractian / Augmentir — document analysis, AI SOPs, connected-worker capture (heavier implementation)

Wedge for MIRA

Pros

- Standalone: no CMMS rip-and-replace
- Honest ABSTAIN when manuals don’t cover it (Answer Radar discipline)
- Equipment-notebook tenancy + citation SSE already in Hub
- OEM-agnostic plant mix (Rockwell + Siemens + Schneider + Omron freezes prove demand breadth)
- Mobile-first technician path already the locked SUT

Cons

- Competitors have brand + installed CMMS bases
- Manual CONNECT coverage is still the pilot bottleneck
- Guided trees and nameplate ID exist but are dark

> The key thing we want to solve is the loss of knowledge, what we call the tribal knowledge… We also want consistency in troubleshooting, so there’s a standard approach to a problem regardless of who is working on it.
>
> — Li Wang, Rockwell Automation Singapore plant director (Microsoft Source, Sep 24, 2026)
>
> I’ve had some success uploading equipment manuals… it often hallucinates if I just ask a general question… But if I upload the manual I want it to reference it does a good job.
>
> — r/PLC technician on uploading manuals to AI (Sep 2025)

Evidence library

1. MaintainX — State of Industrial Maintenance 2026
2. MaintainX 2026 report PDF
3. Deloitte / Manufacturing Institute — 3.8M jobs / 1.9M unfilled (2024-04-03)

4. r/PLC — Do you use AI in your job?
5. r/PLC — The old manuals were so much better
6. r/PLC — AI helper? (NotebookLM citation tip)
7. r/IndustrialMaintenance — Zero documentation
8. PLCtalk — Documenting Projects
9. Squint — Maintenance & Uptime
10. Siemens Asset Essentials — Maintenance Copilot
11. Rockwell Singapore Maintenance Copilot (Microsoft Source) (2026-09-24)
12. MIRA CAPABILITY_CLOSURE.yaml
13. MIRA equipment-notebook chat route

Complaint survey

Method (not a statistical sample). Counted only first-person or firsthand public complaints that match each pain's inclusion criteria, with a live URL and a ≤25-word verbatim quote. Sources searched: Reddit (r/PLC, r/IndustrialMaintenance, related), control.com, and X public posts. PLCtalk / Mike Holt hits were dropped when URLs returned 403. Bias: English, forum-active technicians, Reddit-heavy — **not a market census**.

Survey date: Oct 31, 2026 (America/New_York). Log file: `docs/market/complaint-survey.csv` (box: `/workspace/hair-on-fire-demand-map/complaint-survey.csv`).

Total counted complaints: 45

#	Pain	Counted	Inclusion (short)
1	Cited answers from OUR manuals at the machine	8	Hunting OEM manuals / Ctrl+F hell / needs their manual for trustworthy answers
2	Junior / night shift as capable as the retiring senior	9	Tribal knowledge in seniors' heads; walk-out on retirement; night-shift silos
3	Cut recovery time when unplanned downtime hits	4	Diagnosis/recovery time during unplanned downtime; line-down urgency
4	Branching troubleshooting that isn't a useless error-code echo	4	Fault codes / guides that restate the fault or dead-end to OEM
5	Photo the nameplate → right manual → ask	4	Cannot ID asset / multi-OEM / missing nameplate → wrong or no manual
6	AI that works without ripping out (or depending on) the CMMS	7	CMMS too complex/hostile for techs; floor still needs non-CMMS answers
7	Record the fix once; next shift gets the guided workflow	5	Shift handoff / past-fix capture / seniors resist documenting

#	Pain	Counted	Inclusion (short)
8	Answers when the help file needs internet (or the cell drops)	4	Online-only help, wrong-system help, or docs stranded across the plant

Top proof quotes

> All knowledge is "tribal" (in the heads of senior techs)
> — r/IndustrialMaintenance — Zero documentation
>
> Fuck you here's a 300 page manual. Find it yourself.
> — r/PLC — The old manuals were so much better
>
> troubleshooting guide… says "there is a problem with the breaker" with no info
> — r/PLC — industry meme thread
>
> We currently use Fiix but it’s too complicated for many of my technicians.
> — r/IndustrialMaintenance — MaintainX vs Limble
>
> help files now require an internet connection to work
> — r/PLC — industry meme thread

What this is not

This map does not authorize merge, deploy, Doppler, networking, PLC, paid cloud agents, or customer-facing launch. Those stay Mike-only. Groq/Together are retired — not blockers.

Next decision for Mike: greenlight FINISH of the cited-manual pilot SKU (and which three CONNECT manual packs), or redirect.